Calculus — A Four-Week Course

Every lecture (short version) plus sample homework and exam problems with an answer key.

Week 1 — Limits and derivatives

The tangent and velocity problems; limits; computing limits; continuity; the derivative; differentiation rules; derivatives of trigonometric and exponential functions.

1.1 The tangent and velocity problems

The two questions that created calculus.

The tangent and velocity problems

Calculus began with two problems that turn out to be the same problem. The **tangent problem** asks for the slope of a curve at a single point. The **velocity problem** asks for the speed of a moving object at a single instant. Algebra cannot answer either directly, because a slope needs two points and a speed needs two times.

Secant lines and average velocity

Between two points on the graph of f, the secant slope is the average rate of change:

$$m_{sec} = [f(x + h) - f(x)] / h.$$

If f(t) is position, this is average velocity over a time interval of length h.

From average to instantaneous

Shrink h. The secant lines swing toward a limiting line — the **tangent line** — and the average velocities settle toward a limiting value — the **instantaneous velocity**. For $s(t) = 16t^2$, the average velocity on [2, 2 + h] is 64 + 16h, which approaches 64 as h approaches 0. Making "approaches" precise is the job of the limit.

A real example

Around 450 BCE Zeno of Elea posed the arrow paradox: at any single instant a flying arrow occupies a space exactly its own size, so at that instant it is not moving — and since time is made of instants, it never moves. Calculus answers Zeno by defining velocity at an instant not from the instant alone but as the limit of average velocities over shrinking intervals around it. Newton and Leibniz each built the method in the late seventeenth century.

1.2 Limits

What a function approaches, whether or not it gets there.

Limits

The **limit** of f(x) as x approaches a is the value f(x) gets arbitrarily close to when x is sufficiently close to a — but not equal to a. The value f(a) itself, or whether f is even defined at a, is irrelevant.

Notation and one-sided limits

$$\lim_{x \to a} f(x) = L.$$

The **one-sided limits** $\lim_{x \to a^-}$ and $\lim_{x \to a^+}$ approach from the left and right. The two-sided limit exists exactly when both one-sided limits exist and are equal. If f(x) grows without bound, we write lim f(x) = ∞ and the graph has a vertical asymptote.

The precise definition

Intuition is not enough for proofs. The **ε–δ definition** says $\lim_{x \to a} f(x) = L$ if for every ε > 0 there is a δ > 0 such that

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

However tight a tolerance ε you demand on the output, some tolerance δ on the input guarantees it.

A real example

In 1734 Bishop George Berkeley attacked Newton's calculus in The Analyst, mocking its vanishing increments as "ghosts of departed quantities": they were treated as nonzero when dividing and as zero when discarded. The objection was sound, and it took more than a century to answer. Augustin-Louis Cauchy in the 1820s and Karl Weierstrass later in the century rebuilt calculus on the limit, and the ε–δ definition is Weierstrass's answer to Berkeley.

1.3 Computing limits

The limit laws, algebraic tricks, and the squeeze theorem.

Computing limits

Limits obey arithmetic. If lim f(x) and lim g(x) both exist, the limit of a sum, difference, product, or quotient is the sum, difference, product, or quotient of the limits — provided no denominator limit is zero.

Direct substitution and indeterminate forms

For polynomials, rational functions (away from zeros of the denominator), roots, and the other elementary functions, the limit is found by substituting x = a. When substitution gives 0/0, the limit is **indeterminate**: factor and cancel, rationalize, or simplify first.

$$\lim_{x \to 3} (x^2 - 9)/(x - 3) = \lim_{x \to 3} (x + 3) = 6.$$

The squeeze theorem

If g(x) ≤ f(x) ≤ h(x) near a and both g and h approach L, then f approaches L too. The squeeze proves the most important trigonometric limit:

$$\lim_{x \to 0} \sin x / x = 1.$$

Limits at infinity are handled by dividing by the highest power: the ratio of leading terms decides a rational function's horizontal asymptote.

A real example

Archimedes, around 250 BCE, trapped π between the perimeters of polygons inscribed in and circumscribed about a circle. Doubling the number of sides from 6 up to 96, he proved 3 10/71 < π < 3 1/7. The circle's perimeter is squeezed between two sequences closing in on the same value — the squeeze theorem, two thousand years before it had a name.

1.4 Continuity

Functions whose graphs have no breaks.

Continuity

A function is **continuous** at a if its limit there equals its value there:

The definition

$$\lim_{x \to a} f(x) = f(a).$$

Three things are required: f(a) is defined, the limit exists, and the two agree. Discontinuities are **removable** (a hole), **jump** (unequal one-sided limits), or **infinite** (a vertical asymptote). Polynomials, rational functions on their domains, roots, exponentials, logarithms, and trigonometric functions are continuous wherever they are defined, as are sums, products, quotients, and compositions of them.

The intermediate value theorem

If f is continuous on [a, b] and N lies between f(a) and f(b), then f(c) = N for some c in (a, b). A continuous function cannot skip values. In particular, a continuous function that changes sign must have a zero — the basis of the **bisection method**, which halves the search interval at each step.

$$f(a) < 0 < f(b) \Rightarrow f(c) = 0 \text{ for some } c.$$

A real example

At every moment there are two diametrically opposite points on Earth's equator with exactly the same temperature. Let T(θ) be the temperature at longitude θ and let g(θ) = T(θ) − T(θ + 180°). Then g at the opposite point is −g, so g changes sign as θ runs halfway around; by the intermediate value theorem it is zero somewhere. Continuity alone forces the result, with no measurement at all.

1.5 The derivative

The instantaneous rate of change, defined by a limit.

The derivative

The **derivative** of f at x is the limit of the difference quotient — the slope of the tangent line, the instantaneous rate of change:

The definition

$$f'(x) = \lim_{h \to 0} [f(x + h) - f(x)] / h.$$

Leibniz's notation dy/dx records it as a ratio of changes; Newton's dot notation survives in physics for time derivatives. The **second derivative** f″ is the derivative of f′ — for position, it is acceleration.

Differentiability

A function differentiable at a is continuous there, but not conversely. A derivative fails to exist at a **corner** (|x| at 0), a **cusp**, a **vertical tangent** (the cube root at 0), or any discontinuity. The tangent line at a is

$$y = f(a) + f'(a)(x - a).$$

A real example

Mathematicians long assumed continuous functions must be differentiable at most points. In 1872 Karl Weierstrass presented a function — an infinite sum of cosine waves of increasing frequency and decreasing height — that is continuous everywhere and differentiable nowhere. Its graph is all corners. The shock established that continuity and differentiability are genuinely different, and such functions later became the mathematics of fractals and Brownian motion.

1.6 Differentiation rules

Computing derivatives without limits.

Differentiation rules

Applying the limit definition every time is impractical. A handful of rules, each proved once from the definition, differentiate every algebraic function.

Power, sum, and constant rules

$$d/dx\ x^n = n\ x^{n-1}.$$

The power rule holds for every real exponent n. The derivative of a constant is 0, constants factor out, and the derivative of a sum is the sum of the derivatives.

Product and quotient rules

The derivative of a product is **not** the product of the derivatives:

$$(fg)' = f'g + fg',\ (f/g)' = (f'g - fg') / g^2.$$

The quotient rule follows from the product rule applied to f · (1/g). Higher derivatives are found by applying the rules repeatedly.

A real example

Gottfried Wilhelm Leibniz published the first account of the differential calculus in 1684, in a six-page paper in the journal Acta Eruditorum titled "Nova Methodus pro Maximis et Minimis." In it he stated the product rule in his own notation, d(xv) = x dv + v dx, along with the quotient and power rules — and the notation dx, dy that students still use.

1.7 Derivatives of trigonometric and exponential functions

The transcendental functions, and why radians and e are natural.

Derivatives of trigonometric and exponential functions

From the limit sin x / x → 1 and the sum formula for sine, the derivatives of the trigonometric functions follow:

Trigonometric derivatives

$$d/dx\ \sin x = \cos x,\ d/dx\ \cos x = -\sin x.$$

The quotient rule then gives d/dx tan x = $\sec^2 x$ and the rest. These formulas hold only in radians; in degrees each would carry a factor of π/180.

Exponential derivatives

Differentiating b^x from the definition gives b^x times a constant that depends on b. The number e is the unique base for which that constant is 1:

$$d/dx\ e^x = e^x,\ d/dx\ b^x = b^x \ln b.$$

The exponential e^x is its own derivative — the reason it describes every quantity whose rate of change is proportional to its size.

A real example

A mass on a spring oscillates as x(t) = A cos ωt. Its velocity x′(t) = −Aω sin ωt is zero at the extremes of the motion and greatest as it passes through equilibrium; its acceleration x″(t) = $-\omega^2$ x is always proportional and opposite to displacement. Christiaan Huygens exploited this regular oscillation in the pendulum clock he built in 1656, which improved timekeeping accuracy from minutes to seconds per day.

Week 2 — Differentiation techniques and applications

The chain rule; implicit differentiation and inverse functions; related rates; linear approximation and Newton's method; extreme values and the mean value theorem; curve sketching; optimization.

2.1 The chain rule

Differentiating compositions — the most used rule in calculus.

The chain rule

Most functions are compositions: $\sin(x^2)$, e^{3x}, $\sqrt{(1 + x^4)}$. The **chain rule** differentiates them by multiplying rates of change.

The rule

$$d/dx\ f(g(x)) = f'(g(x)) \cdot g'(x).$$

In Leibniz notation, if y depends on u and u depends on x,

$$dy/dx = (dy/du)(du/dx).$$

Differentiate the outer function leaving the inner alone, then multiply by the derivative of the inner function.

Longer chains

For compositions of three or more functions the rule repeats: each layer contributes a factor. Thus $d/dx\ \sin^3(2x) = 3\sin^2(2x) \cdot \cos(2x) \cdot 2$. Nearly every derivative beyond the basic rules uses the chain rule, and its reverse — substitution — is the main integration technique of Week 3.

A real example

Near sea level, air pressure falls about 1 inch of mercury for every 1,000 feet of altitude. An airliner climbing at 2,000 feet per minute therefore sees cabin-outside pressure change at $dP/dt = (dP/dh)(dh/dt) \approx (-1/1{,}000)(2{,}000) = -2$ inches of mercury per minute. Pressurization systems are designed around exactly this product of rates.

2.2 Implicit differentiation and inverse functions

Differentiating equations that are not solved for y.

Implicit differentiation and inverse functions

Curves such as $x^2 + y^2 = 25$ define y only implicitly. **Implicit differentiation** differentiates both sides with respect to x, treating y as a function of x and applying the chain rule to every y term, then solves for dy/dx.

The method

From $x^2 + y^2 = 25$: $2x + 2y\,y' = 0$, so

$$dy/dx = -x / y.$$

At (3, 4) the tangent slope is −3/4, perpendicular to the radius, as geometry requires.

Inverse functions and logarithms

Differentiating $f(f^{-1}(x)) = x$ gives the derivative of any inverse, and in particular:

$$d/dx \ln x = 1/x,\ d/dx \arctan x = 1/(1 + x^2).$$

Logarithmic differentiation — take ln of both sides, then differentiate implicitly — handles functions like x^x whose variable appears in both base and exponent.

A real example

In 1638 René Descartes challenged Pierre de Fermat to find the tangent lines to the curve $x^3 + y^3 = 3axy$, now called the folium of Descartes, expecting Fermat's method to fail. Fermat solved it. Implicit differentiation now does it in two lines: $3x^2 + 3y^2y' = 3a(y + xy')$, so $y' = (ay - x^2)/(y^2 - ax)$.

2.3 Related rates

When two changing quantities are linked by an equation.

Related rates

If two quantities are related by an equation and both change over time, their rates of change are related by the derivative of that equation with respect to time.

The procedure

- Draw a diagram and name every changing quantity as a function of t.
- Write an equation relating them — geometry, the Pythagorean theorem, similar triangles.
- Differentiate both sides with respect to t using the chain rule.
- Only then substitute the numbers for the instant in question.

$$x^2 + y^2 = z^2 \Rightarrow x\,x' + y\,y' = z\,z'.$$

The classic error

Substituting values before differentiating turns variables into constants and destroys their rates. A ladder's length is constant, so its rate is zero; the distances of its foot and top are not, so they must remain variables until after differentiation.

A real example

A police radar gun measures the rate at which the distance to a car is changing, not the car's speed. From $z^2 = x^2 + d^2$, with d the officer's distance from the road, $z' = (x/z)x' = v \cos\theta$. An officer standing off to the side therefore always reads low — the "cosine effect" — and the error grows as the car approaches. The discrepancy always favors the driver.

2.4 Linear approximation and Newton's method

Using the tangent line to estimate and to solve.

Linear approximation and Newton's method

Near a point where f is differentiable, its graph is nearly straight. The tangent line — the **linearization** — approximates f nearby:

Linear approximation

$$f(x) \approx L(x) = f(a) + f'(a)(x - a).$$

Thus $\sqrt{4.1} \approx 2 + (1/4)(0.1) = 2.025$ (true value 2.0248…). In differential notation, $dy = f'(x)\,dx$ estimates the change in y, and physics uses the same idea whenever it replaces $\sin\theta$ by θ for small angles.

Newton's method

To solve $f(x) = 0$, start at a guess x_n, follow the tangent line to where it crosses the axis, and repeat:

$$x_{n+1} = x_n - f(x_n) / f'(x_n).$$

Near a simple root the number of correct digits roughly doubles with each step. The method can fail when f' is near zero or the starting guess is poor.

A real example

The 1999 video game Quake III Arena needed $1/\sqrt{x}$ millions of times a second to light its 3D scenes. Its source code computed a rough initial guess by manipulating the bits of the floating-point number, then applied a single step of Newton's method to refine it — accurate to within about 0.2 percent and far faster than the hardware's division and square root at the time.

2.5 Extreme values and the mean value theorem

Where maxima and minima can occur, and what derivatives guarantee.

Extreme values and the mean value theorem

The **extreme value theorem** guarantees that a continuous function on a closed interval [a, b] attains both an absolute maximum and an absolute minimum. The question is where.

Critical points

Fermat's theorem: if f has a local extremum at an interior point c and $f'(c)$ exists, then $f'(c) = 0$. So extrema occur only at **critical points** — where f' is zero or undefined — or at endpoints. The **closed interval method** evaluates f at all of them and picks the largest and smallest.

$$f'(c) = 0 \text{ or } f'(c) \text{ undefined}.$$

The mean value theorem

If f is continuous on [a, b] and differentiable on (a, b), then at some interior point the instantaneous rate equals the average rate:

$$f'(c) = [f(b) - f(a)] / (b - a).$$

Consequences: a function with zero derivative is constant, and a function with positive derivative is increasing.

A real example

Average-speed cameras, used on British motorways, photograph a car at two points miles apart and divide the distance by the elapsed time. They never measure instantaneous speed. But if the average exceeds the limit, the mean value theorem guarantees that at some instant between the cameras the speedometer read exactly that average — so the driver was speeding at some moment, whether or not anyone saw it.

2.6 Curve sketching

Reading the shape of a graph from its derivatives.

Curve sketching

The first and second derivatives describe a graph's shape completely enough to sketch it without plotting.

The first derivative test

f is increasing where $f' > 0$ and decreasing where $f' < 0$. At a critical point where f' changes from positive to negative there is a local maximum; from negative to positive, a local minimum; no sign change, no extremum.

Concavity and inflection

f is **concave up** where $f'' > 0$ (bending upward, like a cup) and **concave down** where $f'' < 0$. A point where concavity changes is an **inflection point**. The **second derivative test**:

$$f'(c) = 0,\ f''(c) > 0 \Rightarrow \text{local minimum at } c.$$

A complete sketch combines domain, intercepts, symmetry, asymptotes, intervals of increase, extrema, concavity, and inflection points.

A real example

In the early weeks of the COVID-19 pandemic in 2020, epidemiologists watched cumulative case counts for an inflection point. While cumulative cases are concave up, daily new cases — the derivative — are rising; the inflection point is where daily cases peak and begin to fall. "Flattening the curve" meant reaching that inflection sooner and lower.

2.7 Optimization

Finding the best possible value of a quantity.

Optimization

Optimization problems ask for the largest or smallest value of a quantity — area, cost, time, profit — subject to constraints. The derivative turns them into equation solving.

The procedure

- Identify the quantity Q to optimize and draw a diagram.
- Express Q in terms of one variable, using the constraint to eliminate the others.
- Find the domain, then the critical points where $Q' = 0$.
- Confirm the optimum by the closed interval method or a derivative test.

$$Q'(x) = 0.$$

Economic interpretation

Profit is maximized where marginal revenue equals marginal cost, $R'(x) = C'(x)$: the last unit produced adds exactly as much revenue as cost. A fence of fixed length enclosing the most area is a square; a can holding fixed volume with least material has height equal to its diameter.

A real example

In 1662 Pierre de Fermat proposed that light travels between two points along the path that takes the least time. Minimizing the travel time of a ray crossing from air into water — setting the derivative of total time to zero — yields $\sin\theta_1 / v_1 = \sin\theta_2 / v_2$, exactly Snell's law of refraction. An observed law of optics turned out to be the solution of an optimization problem.

Week 3 — Integration

Antiderivatives; area and Riemann sums; the definite integral; the fundamental theorem of calculus; substitution; integration by parts; numerical integration.

3.1 Antiderivatives

Running differentiation in reverse.

Antiderivatives

An **antiderivative** of f is a function F with F′ = f. Because constants have derivative zero, antiderivatives come in families: if F is one, every other is F + C.

Basic formulas

Each derivative rule read backwards is an antiderivative rule:

$$\int x^n \, dx = x^{n+1} / (n + 1) + C, \; n \neq -1.$$

- $\int 1/x \, dx = \ln|x| + C$ — the case $n = -1$;
- $\int e^x \, dx = e^x + C$;
- $\int \cos x \, dx = \sin x + C$ and $\int \sin x \, dx = -\cos x + C$.

Initial value problems

A condition such as F(0) = 5 selects one member of the family by determining C. Given acceleration, one antiderivative gives velocity and a second gives position, with the initial velocity and initial position as the two constants.

A real example

Near Earth's surface a falling body has constant acceleration of about −32 feet per second squared. Antidifferentiating once gives $v(t) = -32t + v_0$; again gives $s(t) = -16t^2 + v_0 t + s_0$. Galileo discovered from experiment that distance fallen grows with the square of time; calculus derives his law in two lines from the single fact of constant acceleration.

3.2 Area and Riemann sums

Approximating area by rectangles.

Area and Riemann sums

Finding the area under a curve is the second founding problem of calculus. The strategy: slice the region into thin vertical strips, approximate each by a rectangle, and add.

Riemann sums

Divide [a, b] into n subintervals of width $\Delta x = (b - a)/n$ and choose a sample point x_i^* in each:

$$\Sigma_{i=1}^{n} f(x_i^*) \, \Delta x.$$

Left endpoints, right endpoints, and midpoints give different approximations; for an increasing function, left sums underestimate and right sums overestimate.

Sigma notation

Summation formulas make exact limits possible: $\Sigma\, i = n(n+1)/2$ and $\Sigma\, i^2 = n(n+1)(2n+1)/6$. Using them, the right-endpoint sum for x^2 on [0, 1] is $(n+1)(2n+1)/(6n^2)$, which approaches 1/3 as $n \to \infty$. The area is the limit of the sums.

A real example

Around 250 BCE Archimedes found the area of a parabolic segment by filling it with triangles: one large triangle, then two smaller ones in the gaps, then four, and so on, each generation having one-quarter the area of the one before. Summing $1 + 1/4 + 1/16 + \ldots = 4/3$, he proved the segment has exactly four-thirds the area of the first triangle — an integral computed nineteen centuries before calculus.

3.3 The definite integral

The limit of Riemann sums, and what it measures.

The definite integral

Bernhard Riemann gave the general definition in 1854. The **definite integral** of f from a to b is the limit of Riemann sums as the subintervals shrink:

Definition and interpretation

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \Sigma_{i=1}^{n} f(x_i^*)\,\Delta x.$$

It measures **net signed area**: area above the axis counts positive, area below negative. Every continuous function on [a, b] is integrable.

Properties

- The integral is linear: constants factor out, and the integral of a sum is the sum of the integrals.
- $\int_a^b + \int_b^c = \int_a^c$, and reversing the limits changes the sign.
- The **average value** of f on [a, b] is $(1/(b-a)) \int_a^b f(x)\,dx$.

The integral of a rate of change is the net change: integrating velocity gives displacement, not total distance traveled.

A real example

A household electric meter is an integrating machine. Power drawn, P(t), fluctuates second by second as appliances switch on and off; the bill charges for energy, $E = \int P(t)\,dt$, measured in kilowatt-hours. A 1,500-watt heater running for two hours contributes $1.5 \times 2 = 3$ kilowatt-hours — the area of a rectangle under the power graph.

3.4 The fundamental theorem of calculus

Differentiation and integration are inverse processes.

The fundamental theorem of calculus

Riemann sums compute integrals slowly. The **fundamental theorem of calculus** connects the two halves of the subject and turns integration into antidifferentiation.

Part 1

If f is continuous, the accumulation function has f as its derivative:

$$d/dx \int_a^x f(t)\,dt = f(x).$$

The rate at which area accumulates under a curve equals the height of the curve. In particular, every continuous function has an antiderivative.

Part 2

If F is any antiderivative of f on [a, b], then

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

The area under x^2 from 0 to 1 is $1^3/3 - 0 = 1/3$ at once, with no limits of sums.

A real example

Isaac Barrow, Newton's teacher at Cambridge, proved a geometric version of the theorem in the 1660s. But Newton and Leibniz each recognized it as a general computational method, and that recognition is what made calculus a calculus. Areas that had required Archimedes' ingenuity — one clever argument per curve — became routine: find an antiderivative and subtract.

3.5 Substitution

The chain rule run backwards.

Substitution

Many integrands are the derivative of a composition, f(g(x))g′(x). **Substitution** recognizes the pattern and simplifies it by renaming the inner function.

The rule

$$\int f(g(x))\,g'(x)\,dx = \int f(u)\,du,\ u = g(x).$$

Choose u as an inner function whose derivative appears (up to a constant) as a factor. Thus $\int 2x\cos(x^2)\,dx$ becomes $\int \cos u\,du = \sin(x^2) + C$.

Definite integrals

For a definite integral, change the limits along with the variable:

$$\int_a^b f(g(x))\,g'(x)\,dx = \int_{g(a)}^{g(b)} f(u)\,du.$$

Then there is no need to return to x. Substitution also yields $\int \tan x\,dx = -\ln|\cos x| + C$, with u = cos x.

A real example

The Mercator map, published in 1569, spaces lines of latitude so that compass bearings appear as straight lines; the spacing at latitude φ is the integral of sec φ. Edward Wright tabulated it in 1599 by adding up secants numerically. Around 1645 Henry Bond noticed the table matched ln tan(π/4 + φ/2), and calculus later confirmed it: multiplying sec φ by (sec φ + tan φ)/(sec φ + tan φ) sets up a substitution.

3.6 Integration by parts

The product rule run backwards.

Integration by parts

Integrating $(uv)' = u'v + uv'$ and rearranging gives the second major technique, for integrands that are products of unlike functions.

The formula

$$\int u\,dv = uv - \int v\,du.$$

Choose u to become simpler when differentiated and dv to be easy to integrate. For $\int x\,e^x\,dx$, take $u = x$ and $dv = e^x\,dx$: the result is $x\,e^x - e^x + C$.

Patterns

- Polynomial times exponential or trigonometric: let u be the polynomial and repeat as needed.
- Logarithms and inverse trigonometric functions: let u be that function, dv = dx — this gives $\int \ln x\,dx = x\ln x - x + C$.
- $e^x \sin x$: two applications return the original integral, which is then solved for algebraically.

A real example

Integration by parts gives the Laplace transform its central property: the transform of a derivative f' equals s times the transform of f, minus f(0). This turns linear differential equations into algebraic ones. Electrical and control engineers use it daily to analyze circuits, feedback systems, and autopilots — each application a disguised integration by parts.

3.7 Numerical integration

Computing integrals when no antiderivative can be found.

Numerical integration

Many integrals cannot be done by formula. Joseph Liouville proved in the 1830s that e^{-x^2} — the bell curve — has no elementary antiderivative, yet its integrals are needed everywhere in statistics. They are computed numerically.

The trapezoidal rule

Replace the curve on each subinterval by a straight line:

$$T_n = (\Delta x/2)[f(x_0) + 2f(x_1) + \dots + 2f(x_{n-1}) + f(x_n)].$$

The error shrinks in proportion to $1/n^2$: doubling n cuts it by about a factor of four.

Simpson's rule

Replace the curve on each pair of subintervals by a parabola (n even):

$$S_n = (\Delta x/3)[f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 4f(x_{n-1}) + f(x_n)].$$

Its error shrinks in proportion to $1/n^4$, so doubling n cuts it by about sixteen. Simpson's rule is exact for cubics.

A real example

Naval architects compute the areas and volumes of ship hulls from tables of offsets — widths measured at evenly spaced stations along the hull. Simpson's rules have been the standard tool for this for centuries: applied to the offsets they give the waterplane area, the underwater volume, and hence the displacement and stability of a ship before it is built.

Week 4 — Applications of integration and infinite series

Area between curves; volumes; differential equations; improper integrals and L'Hôpital's rule; infinite series; power series; Taylor series; a capstone synthesis.

4.1 Area between curves

Integrating the gap between two functions.

Area between curves

If f(x) ≥ g(x) on [a, b], the area of the region between them is the integral of the vertical gap — top minus bottom:

The formula

$$A = \int_a^b [f(x) - g(x)]\,dx.$$

The limits are usually the x-coordinates where the curves intersect, found by solving f(x) = g(x). If the curves cross inside the interval, split the integral where they cross and integrate |f − g|.

Integrating in y

When a region is bounded on the left and right by curves x = f(y) and x = g(y), slice horizontally instead:

$$A = \int_c^d [\text{right} - \text{left}]\,dy.$$

Choosing the direction of slicing that avoids splitting the region often turns two integrals into one.

A real example

Economists measure income inequality with the **Gini coefficient**. The Lorenz curve L(x) gives the share of total income earned by the poorest fraction x of the population; perfect equality is the line y = x. The Gini coefficient is the area between the two curves divided by the area under the line: G = 1 − 2$\int_0^1$ L(x) dx, ranging from 0 (equality) to 1 (one person has everything).

4.2 Volumes

Slicing solids into disks, washers, and shells.

Volumes

A volume, like an area, is an integral: slice the solid into thin pieces whose volumes are easy, and add. If A(x) is the cross-sectional area at x, V = $\int_a^b$ A(x) dx.

Disks and washers

Rotating y = f(x) about the x-axis gives circular cross-sections of radius f(x):

$$V = \pi \int_a^b [f(x)]^2 \, dx.$$

If the region has a hole, each slice is a **washer** with area $\pi(R^2 - r^2)$.

Cylindrical shells

Rotating about the y-axis, it is often easier to use thin cylindrical shells of radius x and height f(x):

$$V = 2\pi \int_a^b x \, f(x) \, dx.$$

Disks slice perpendicular to the axis; shells slice parallel to it. Choose whichever gives the simpler integral.

A real example

In 1613 Johannes Kepler noticed that Austrian wine merchants priced barrels by pushing a rod diagonally through the bung-hole and reading its length, whatever the barrel's shape. Skeptical, he wrote Nova Stereometria Doliorum Vinariorum (1615), computing volumes of solids of revolution by slicing them into infinitely many thin pieces — an early step toward integral calculus, prompted by a wine purchase.

4.3 Differential equations

Equations whose unknown is a function.

Differential equations

A **differential equation** relates a function to its derivatives. It expresses a law of change; solving it predicts the future from the present.

Exponential growth and decay

The simplest model says the rate of change is proportional to the amount present:

$$dy/dt = ky \Rightarrow y = y_0 \, e^{kt}.$$

It is solved by **separation of variables**: dy/y = k dt, integrate both sides, and exponentiate.

Logistic growth and slope fields

Real populations are limited by a carrying capacity K:

$$dP/dt = rP(1 - P/K).$$

Growth is nearly exponential when P is small and slows to zero as P approaches K, producing the S-shaped logistic curve. A **slope field** sketches solutions graphically when no formula is available; Euler's method follows it numerically in small steps.

A real example

Newton's law of cooling says an object's temperature changes at a rate proportional to its difference from the surroundings: $dT/dt = -k(T - T_a)$. Its solution, $T = T_a + (T_0 - T_a)e^{-kt}$, is used in forensic medicine: from two body-temperature readings taken some time apart, investigators solve for k and then for the time at which the body was at a normal 98.6°F — an estimate of the time of death.

4.4 L'Hôpital's rule and improper integrals

Indeterminate limits, and integrals over infinite intervals.

L'Hôpital's rule and improper integrals

Limits of the form 0/0 or ∞/∞ can be evaluated by differentiating numerator and denominator separately.

L'Hôpital's rule

$$\lim_{x\to a} f(x)/g(x) = \lim_{x\to a} f'(x)/g'(x).$$

It shows that exponentials beat polynomials, which beat logarithms: $\lim x^n/e^x = 0$ and $\lim \ln x / x = 0$. Other indeterminate forms — $0\cdot\infty$, $\infty - \infty$, 1^∞, 0^0 — are first rewritten as quotients. The rule is actually Johann Bernoulli's; the Marquis de L'Hôpital paid for his results and published them in the first calculus textbook in 1696.

Improper integrals

An integral over an infinite interval, or of a function with a vertical asymptote, is defined as a limit:

$$\int_1^\infty f(x)\,dx = \lim_{t\to\infty} \int_1^t f(x)\,dx.$$

It **converges** if the limit is finite. The benchmark: $\int_1^\infty 1/x^p\,dx$ converges exactly when $p > 1$.

A real example

In 1641 Evangelista Torricelli rotated $y = 1/x$ for $x \geq 1$ about the x-axis. The resulting infinitely long horn — now called Gabriel's horn — has volume $\pi\int_1^\infty 1/x^2\,dx = \pi$, yet its surface area is infinite, since it exceeds $2\pi\int_1^\infty 1/x\,dx$. A horn you could fill with paint but never paint was a scandal to seventeenth-century philosophers.

4.5 Infinite series

Adding infinitely many numbers.

Infinite series

An **infinite series** $\Sigma\, a_n$ is the limit of its partial sums $S_N = a_1 + \dots + a_N$. It **converges** if that limit exists and **diverges** otherwise.

Basic results

- **Divergence test**: if a_n does not approach 0, the series diverges. The converse is false.
- **Geometric series**: $\Sigma\, ar^n$ converges to $a/(1 - r)$ when $|r| < 1$.
- **p-series**: $\Sigma\, 1/n^p$ converges exactly when $p > 1$ — the integral test links it to Lecture 4.4.

$$\sum_{n=1}^\infty 1/n = \infty.$$

Convergence tests

The **comparison tests** measure a series against a known one. The **ratio test** examines lim $|a_{n+1}/a_n|$: less than 1 converges, greater than 1 diverges. The **alternating series test** says terms that alternate in sign and decrease to zero give a convergent series, as with $1 - 1/2 + 1/3 - \ldots = \ln 2$.

A real example

Around 1350 Nicole Oresme proved the harmonic series $1 + 1/2 + 1/3 + \ldots$ diverges by grouping its terms into blocks each exceeding 1/2. The series $\Sigma\, 1/n^2$ converges, but to what resisted the best mathematicians of Europe until 1734, when Leonhard Euler showed its sum is $\pi^2/6$ — a result that made his reputation. It is still called the Basel problem.

4.6 Power series

Infinite polynomials.

Power series

A **power series** is a series with a variable: an infinite polynomial centered at a.

Radius of convergence

$$\Sigma_{n=0}^{\infty}\, c_n\,(x-a)^n.$$

Every power series has a **radius of convergence** R: it converges for $|x - a| < R$ and diverges for $|x - a| > R$, with the endpoints checked separately. The ratio test usually finds R. The model example is the geometric series $1/(1 - x) = 1 + x + x^2 + \ldots$, with $R = 1$.

Calculus with power series

Inside its interval of convergence a power series can be differentiated and integrated term by term, like a polynomial. Integrating $1/(1 + x^2) = 1 - x^2 + x^4 - \ldots$ gives

$$\arctan x = x - x^3/3 + x^5/5 - \ldots.$$

Power series thus produce new functions and new representations of old ones.

A real example

Setting $x = 1$ in the arctangent series gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \ldots$. Europeans credit James Gregory and Leibniz, who found it in the 1670s, but Madhava of Sangamagrama and the Kerala school in southern India had it around 1400, along with series for sine and cosine and correction terms to speed convergence — an early mastery of infinite series.

4.7 Taylor series

Every smooth function as a polynomial of infinite degree.

Taylor series

If f has a power series at a, its coefficients are forced by its derivatives: $c_n = f^{(n)}(a)/n!$. The resulting **Taylor series** is

The series

$$f(x) = \Sigma_{n=0}^{\infty} f^{(n)}(a)\,(x-a)^n / n!.$$

Centered at 0 it is called a Maclaurin series. The essential examples:

- $e^x = 1 + x + x^2/2! + x^3/3! + \ldots$, for all x;
- $\sin x = x - x^3/3! + x^5/5! - \ldots$;
- $\cos x = 1 - x^2/2! + x^4/4! - \ldots$.

Taylor polynomials and error

Truncating after degree n gives the Taylor polynomial T_n, the best polynomial approximation of f near a. Taylor's inequality bounds the error by $M|x-a|^{n+1}/(n+1)!$, where M bounds $|f^{(n+1)}|$. Substituting ix into e^x and comparing with the sine and cosine series gives Euler's formula $e^{ix} = \cos x + i \sin x$.

A real example

Einstein's relativistic kinetic energy is $mc^2(1/\sqrt{(1 - v^2/c^2)} - 1)$. Expanding in a Taylor series in v/c gives $\tfrac{1}{2}mv^2 + (3/8)mv^4/c^2 + \ldots$. The first term is Newton's kinetic energy; the rest are corrections too small to notice at everyday speeds. Taylor series show precisely why Newtonian physics works so well — and where it stops working.

4.8 Capstone synthesis

One idea behind everything: the local determines the global.

Capstone synthesis

Step back, and the entire course reduces to a single sentence: calculus studies change by taking limits, and the derivative and the integral are inverse operations.

The one theorem

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

Everything you have learned elaborates this statement:

- Limits make instantaneous rates precise (Week 1).
- Derivatives measure local change and locate optima (Week 2).
- Integrals add up local change into global totals (Week 3).
- Differential equations and series predict and approximate from local information (Week 4).

Where calculus goes next

Multivariable calculus extends these ideas to functions of several variables — partial derivatives, multiple integrals, and the theorems of Green, Gauss, and Stokes, which generalize the fundamental theorem. Differential equations, probability, and physics rest on everything here.

A real example

Konstantin Tsiolkovsky derived the rocket equation in 1903. Conservation of momentum gives a differential relation: a rocket ejecting mass dm at exhaust speed v_e gains speed dv = $-v_e$ dm/m. Integrating from initial mass m_0 to final mass m_f gives $\Delta v = v_e \ln(m_0/m_f)$. Rates, a differential equation, an integral, and a logarithm — the whole course — explain why rockets are mostly fuel.

Practice problems

A sample of the graded work in the course: two homeworks plus the midterm and final. Answers follow at the end.

Homework 1.1 — Limits and the derivative

Short-answer problems on limits and derivatives. Write each answer as the key formula in symbols using the on-screen math keyboard (use the Calculus and Trigonometry tabs for limits, primes, and Greek letters).

1. Write the limit definition of the derivative f′(x).
2. Write the condition for f to be continuous at x = a.
3. Write the fundamental trigonometric limit as x → 0.
4. Write the power rule for d/dx x^n.

Homework 1.2 — Differentiation rules

Use the math keyboard for fractions, primes, and exponents.

1. Write the product rule for (fg)′.
2. Write the quotient rule for (f/g)′.
3. Write the derivative of sin x.
4. Write the derivative of e^x.

Midterm — Weeks 1 & 2

Cumulative midterm on limits, derivatives, and applications of differentiation. 60 minutes. Math keyboard available; pasting disabled.

1. Write the limit definition of the derivative f′(x).
2. Write the product rule for (fg)′.
3. Write the quotient rule for (f/g)′.
4. Write the chain rule for d/dx f(g(x)).
5. Write the derivative of ln x.
6. Write the linearization L(x) of f at x = a.
7. Write the Newton's method iteration for x_(n+1).
8. Write the conclusion of the mean value theorem on [a, b].

Final Exam — Calculus

Cumulative final covering all four weeks. 90 minutes. Math keyboard available; pasting disabled.

1. Write the limit definition of the derivative f′(x).
2. Write the chain rule for d/dx f(g(x)).
3. Write the power rule for ∫ x^n dx, n ≠ −1.
4. Write the fundamental theorem of calculus, Part 1.
5. Write the fundamental theorem of calculus, Part 2, for an antiderivative F.
6. Write the integration by parts formula.
7. Write the disk-method volume for y = f(x) rotated about the x-axis on [a, b].
8. Write the solution of dy/dt = ky with initial value y_0.
9. Write L'Hôpital's rule for lim f(x)/g(x) as x → a.
10. Write the Taylor series of f centered at a.

Answer key

Homework 1.1 — Limits and the derivative:

1. f′(x) = lim_(h→0) [f(x + h) − f(x)] / h
2. lim_(x→a) f(x) = f(a)
3. lim_(x→0) sin x / x = 1
4. d/dx x^n = n x^(n−1)

Homework 1.2 — Differentiation rules:

1. (fg)′ = f′g + fg′
2. (f/g)′ = (f′g − fg′) / g²
3. d/dx sin x = cos x
4. d/dx e^x = e^x

Midterm — Weeks 1 & 2:

1. f′(x) = lim_(h→0) [f(x + h) − f(x)] / h
2. (fg)′ = f′g + fg′
3. (f/g)′ = (f′g − fg′) / g²
4. d/dx f(g(x)) = f′(g(x)) · g′(x)
5. d/dx ln x = 1/x
6. L(x) = f(a) + f′(a)(x − a)
7. x_(n+1) = x_n − f(x_n) / f′(x_n)
8. f′(c) = [f(b) − f(a)] / (b − a)

Final Exam — Calculus:

1. f′(x) = lim_(h→0) [f(x + h) − f(x)] / h
2. d/dx f(g(x)) = f′(g(x)) · g′(x)
3. ∫ x^n dx = x^(n+1) / (n + 1) + C
4. d/dx ∫_a^x f(t) dt = f(x)
5. ∫_a^b f(x) dx = F(b) − F(a)
6. ∫ u dv = uv − ∫ v du
7. V = π ∫_a^b [f(x)]² dx
8. y = y_0 e^(kt)
9. lim_(x→a) f(x)/g(x) = lim_(x→a) f′(x)/g′(x)
10. f(x) = Σ_(n=0)^∞ f^(n)(a) (x − a)^n / n!